

BTL Industries JSC	702 – Skincare	RELEASE
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	Test report
PROJECT NAME: 702 – Skincare	
FILE NAME: 702-51gener104 Test plan	RELEASE

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1. Scope

The purpose of this document is to present the results from the experiments done while testing the device's functionality.

2. References

- Pumpcontrol102 Test Plan – 702-pumpcontrol102_Test_Plan.docx

3. Equipment

- Oscilloscope – Tektronix MDO3024;
- Multimeter – Fluke 87V;
- IR imager – Agilent U5855A;
- DC power supply – Keysight U8031A.

4. Test conditions

The test conditions are valid for all tests unless otherwise specified.

- Ambient temperature - $23^{\circ}\text{C} \pm 3^{\circ}\text{C}$
- Relative humidity - $30 \div 60 \%$
- Atmospheric pressure - $700 \div 1060 \text{ hPa}$

5. Visual inspection

Table 1. Visual inspection results

Serial number	200182100126	200182100128 (reworked)
Labels	Pass	Pass
Solder joints	Pass	Pass
Components	Pass	Pass
Short circuit	Pass	Pass
PCB solder mask	Pass	Pass

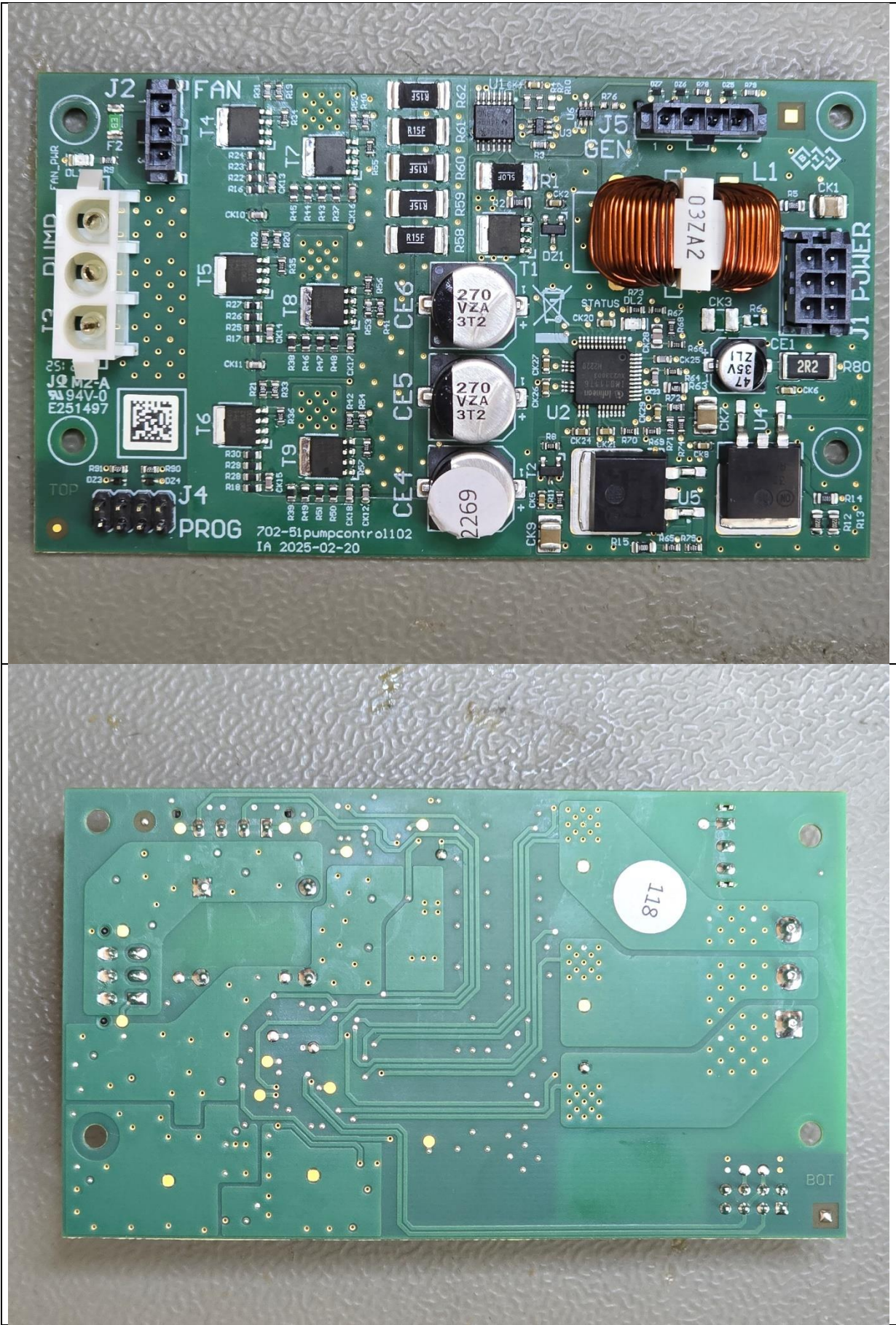


Figure 1. Pictures of 200182100126 - top and bottom sides

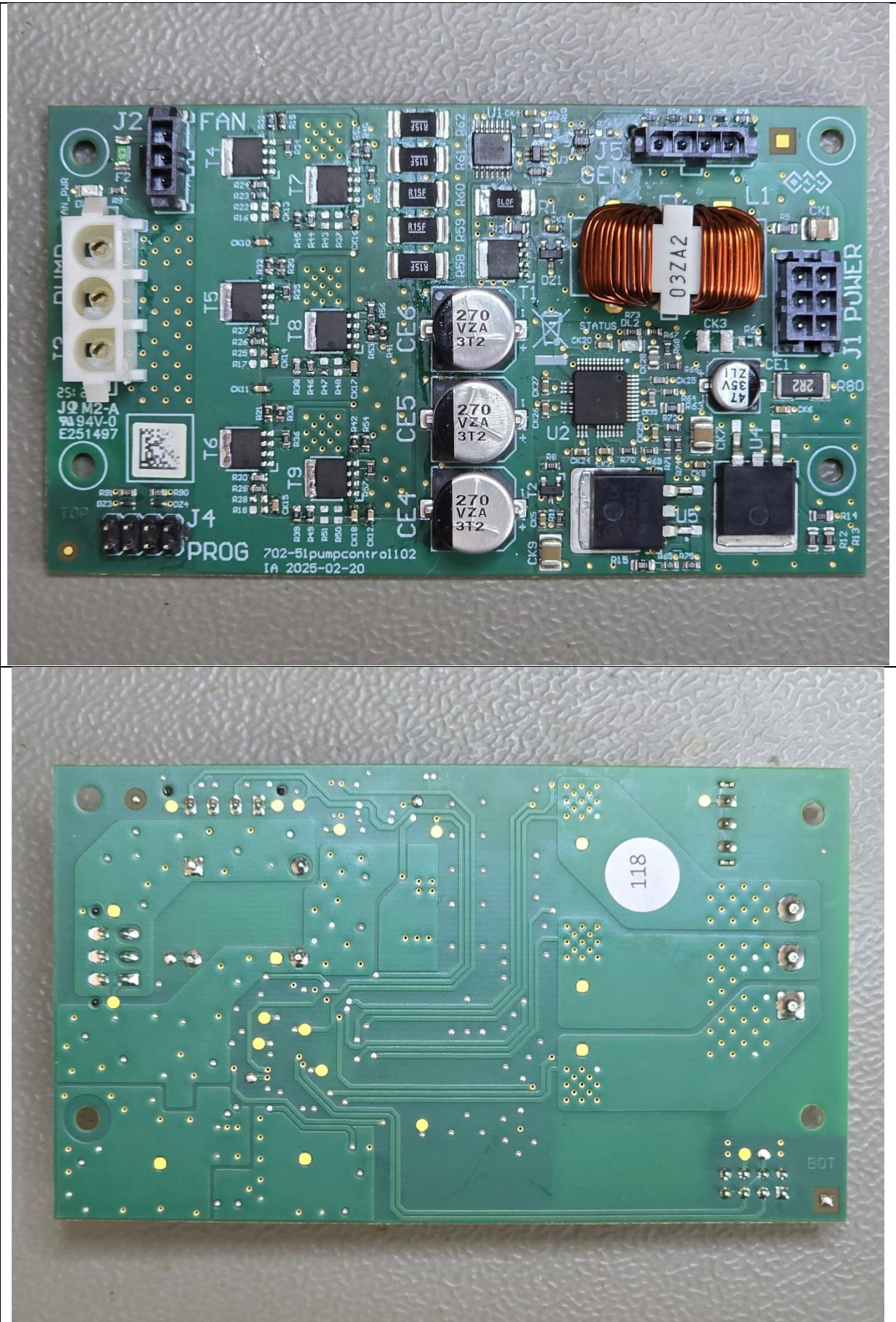


Figure 2. Pictures of 200182100128 (reworked) - top and bottom sides

6. Electrical measurements

6.1. Power



Figure 3. Power test setup

Table 2. Power measurements

Serial number	200182100126	200182100128 (reworked)
Stage 1: Pump/Fan OFF		
U _{+24V_FILT} [V] (TP3)	23.99	24.00
U _{+15V} [V] (TP19)	14.85	14.77
U _{+3.3V} [V] (TP11)	3.315	3.314
U _{UVEN} [V] (TP15)	0.002	0.002
U _{+24V} [V] (TP1)	1.810	1.838
DL1 [ON/OFF]	OFF	OFF
U _{+24V_FAN} [V] (TP4)	1.809	1.837
U _{PG} [V] (TP16)	3.286	3.284
Stage 2: Pump/Fan ON		
U _{UVEN} [V] (TP15)	3.320	3.319
U _{+24V} [V] (TP1)	23.99	23.99
DL1 [ON/OFF]	ON	ON
U _{+24V_FAN} [V] (TP4)	23.87	23.89
U _{PG} [V] (TP16)	0.008	0.009

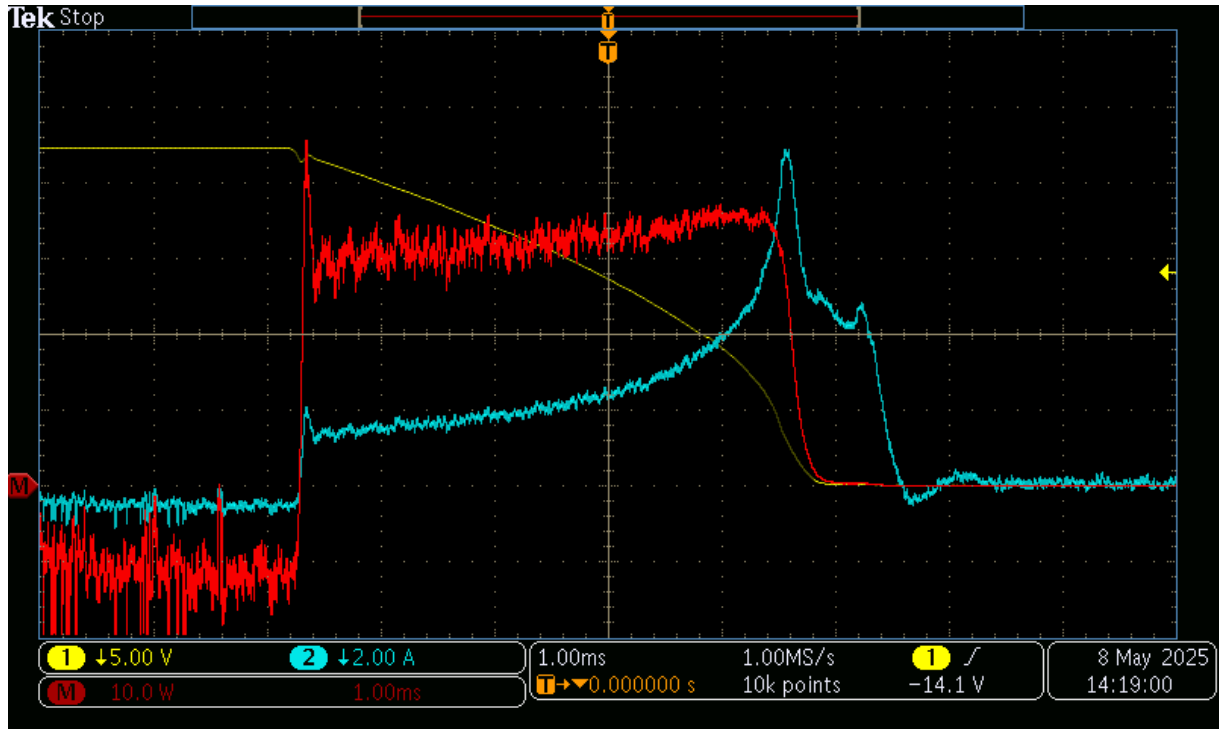


Figure 4. T1 power-on transient (yellow - I_D , blue - U_{DS} , red - $I_D \cdot U_{DS}$) – 200182100126

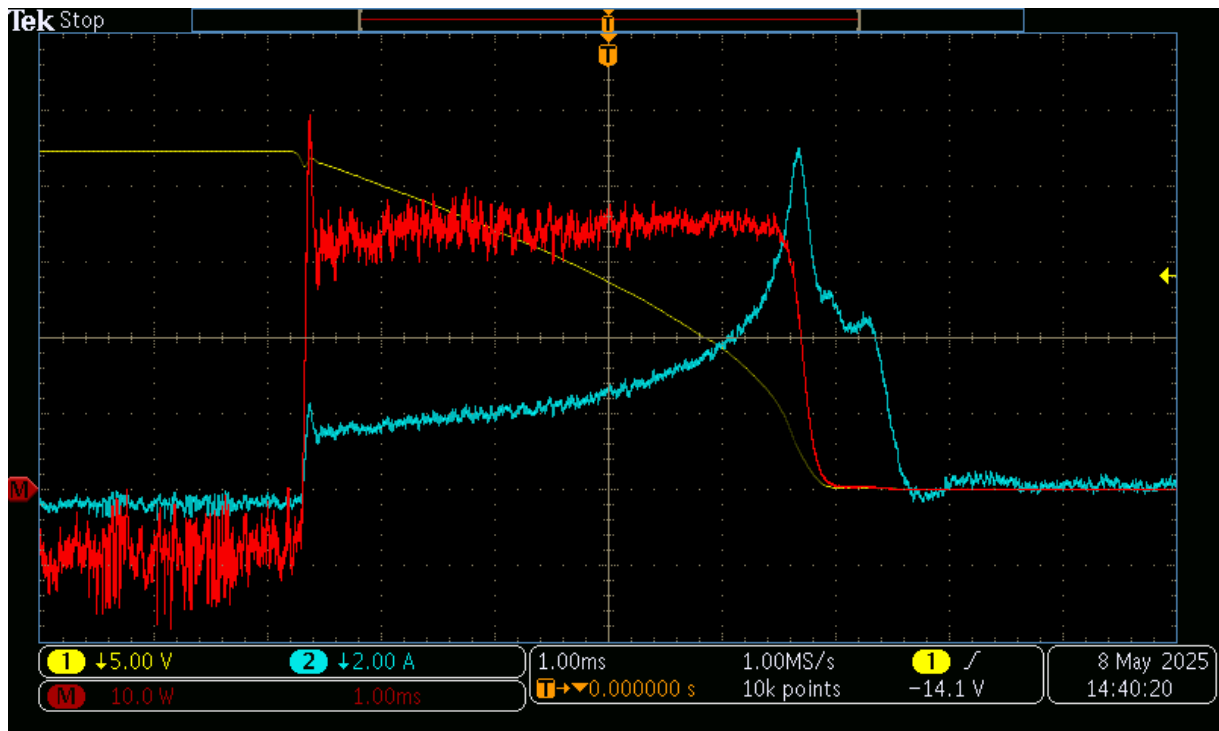


Figure 5. T1 power-on transient (yellow - I_D , blue - U_{DS} , red - $I_D \cdot U_{DS}$) - 200182100128 (reworked)

6.2. Driver



Figure 6. Driver test setup

Table 3. Driver measurements

Serial number	200182100126	200182100128 (reworked)
Stage 1: Pump/Fan OFF		
DL2 [ON/OFF]	ON	ON
U _{BUS_SENSE} [V] (TP17)	0.145	0.148
U _{VSP} [V] (TP22)	0.000	0.000
Stage 2: Pump/Fan ON		
U _{BUS_SENSE} [V] (TP17)	2.116	2.117
U _{VSP} [V] (TP22)	2.786	2.783
f _{SPEED_FB} [Hz] (TP10)	339.5	340.8

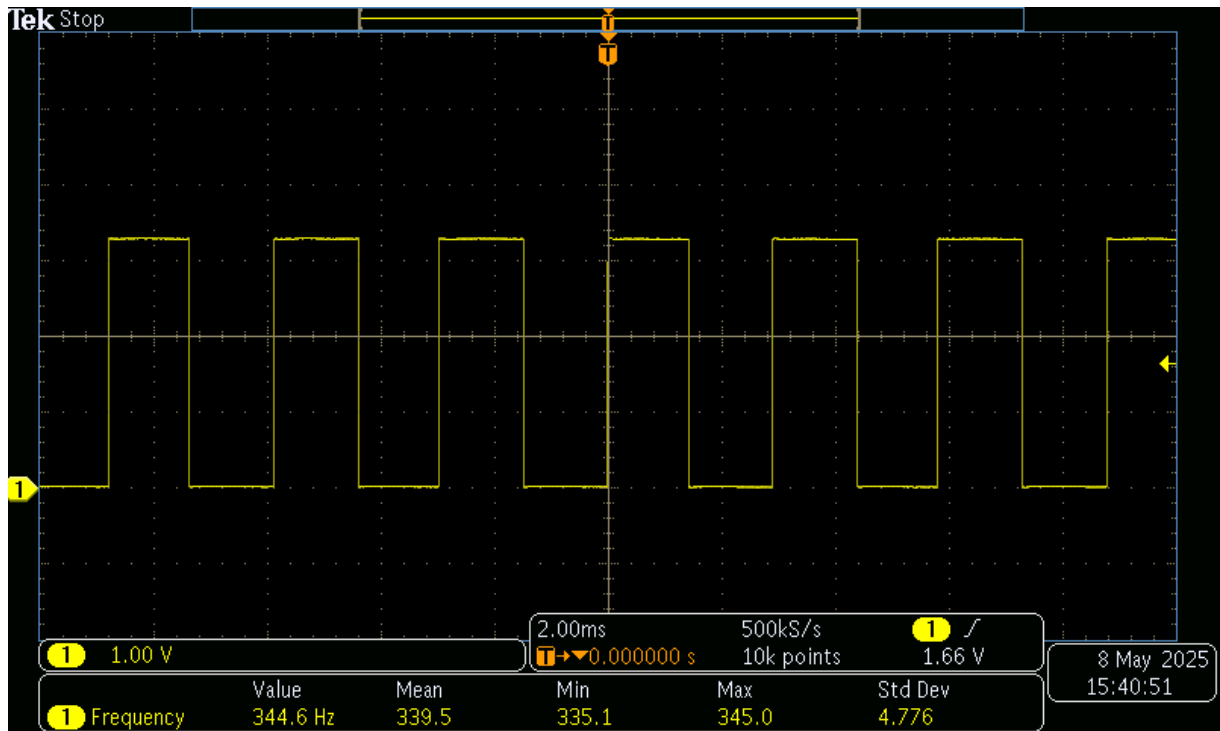


Figure 7. PWM at SPEED_FB – 200182100126

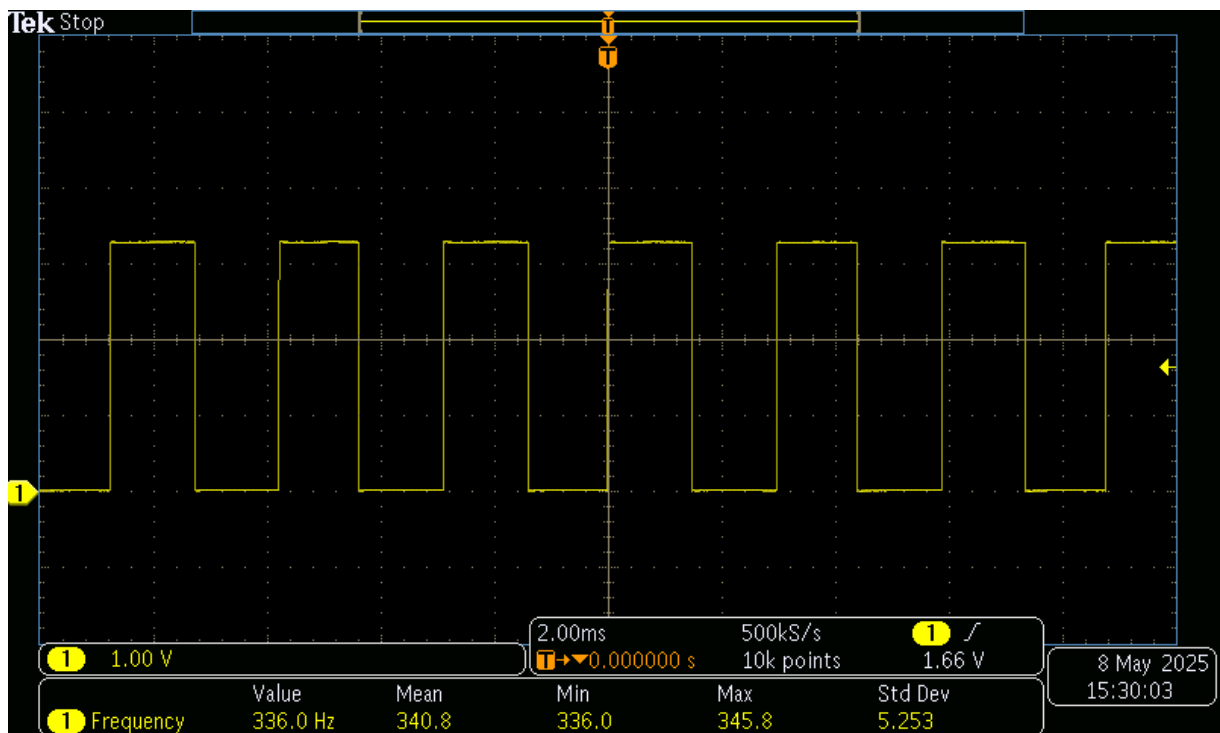


Figure 8. PWM at SPEED_FB – 200182100128 (reworked)

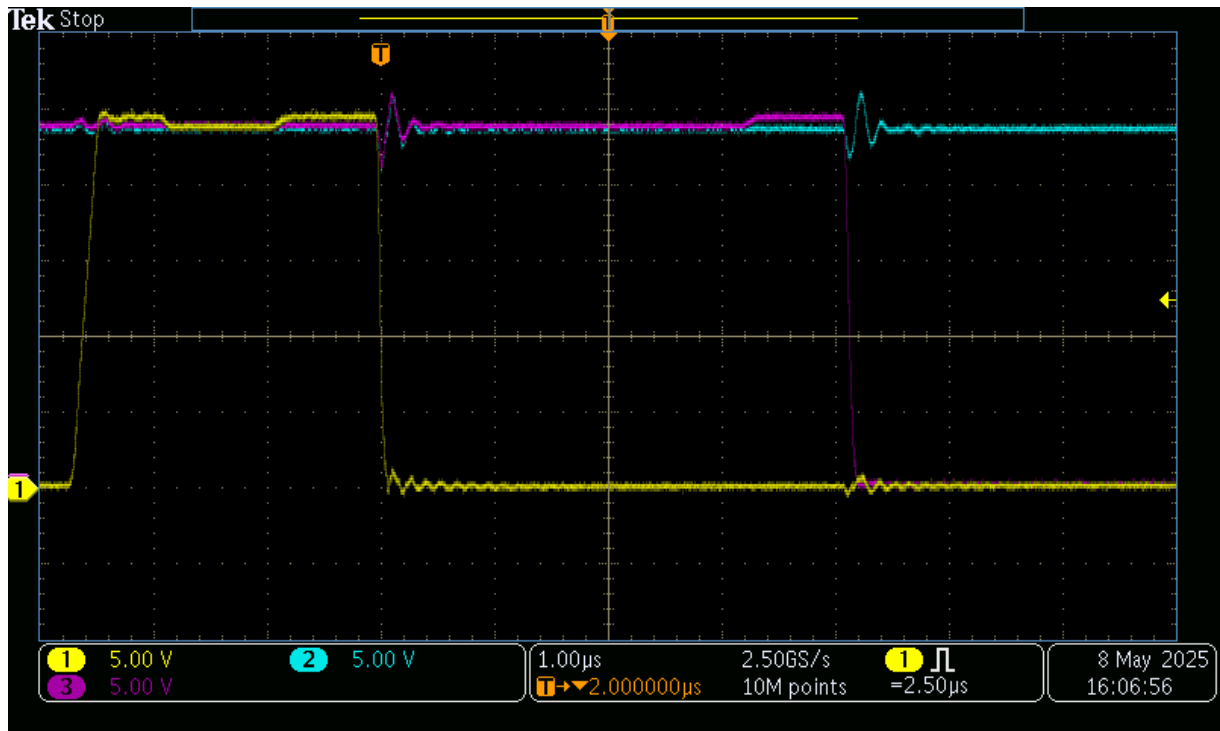


Figure 9. Voltage waveforms at U (yellow), V (blue), W (purple) - 200182100126

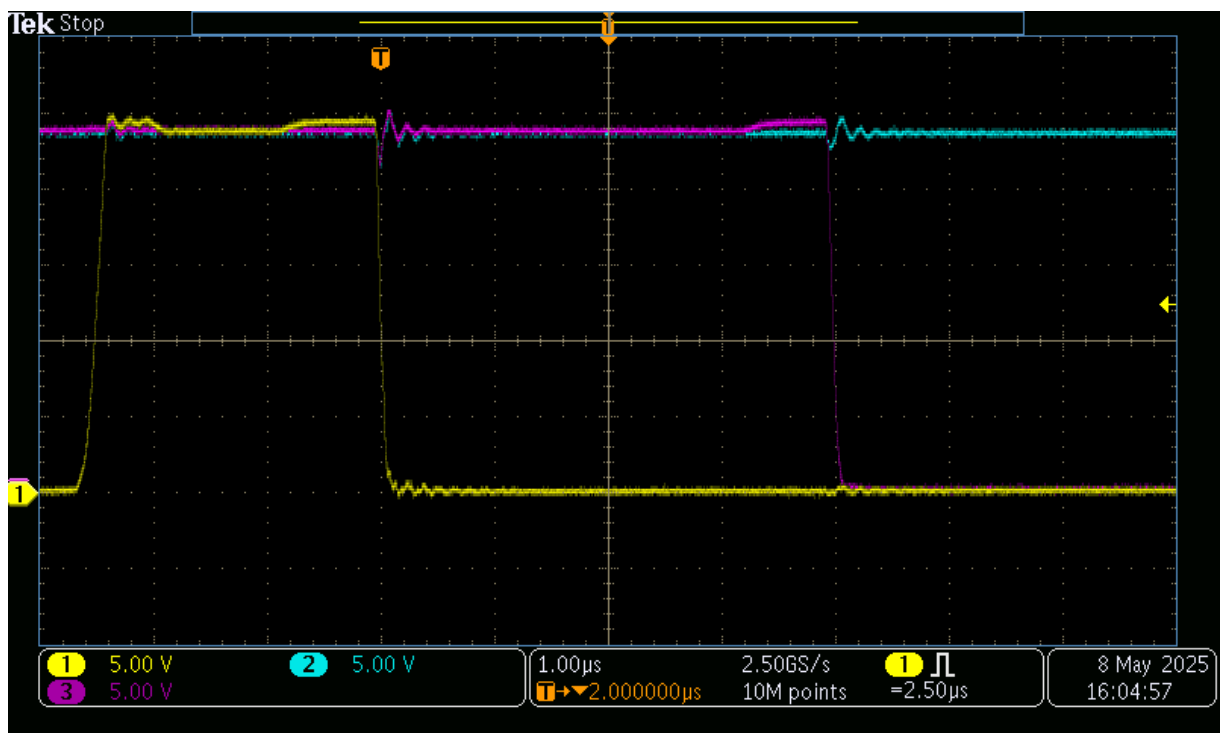


Figure 10. Voltage waveforms at U (yellow), V (blue), W (purple) - 200182100128 (reworked)

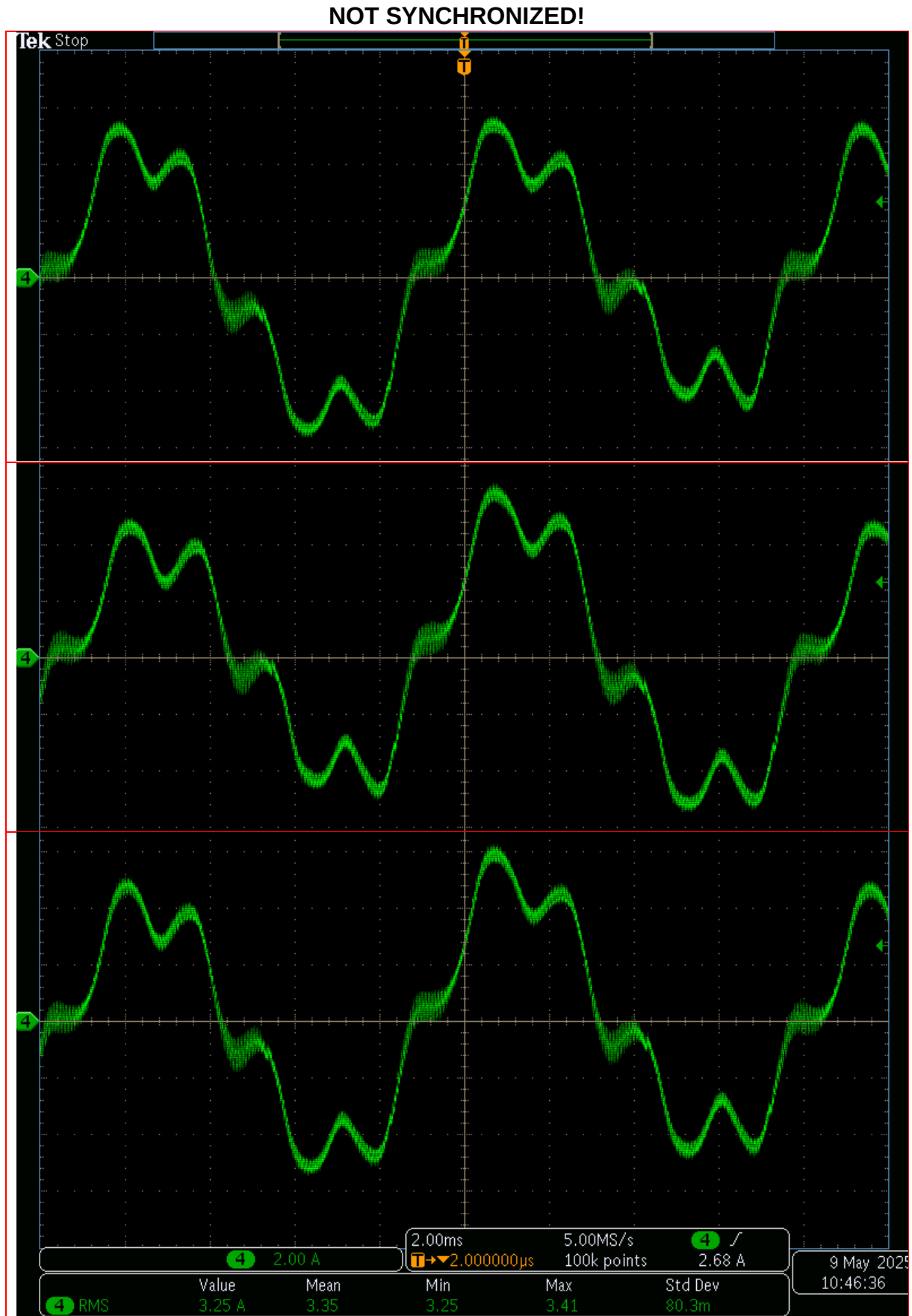


Figure 11. Current waveforms at U (top), V (middle), W (bottom) - 200182100126

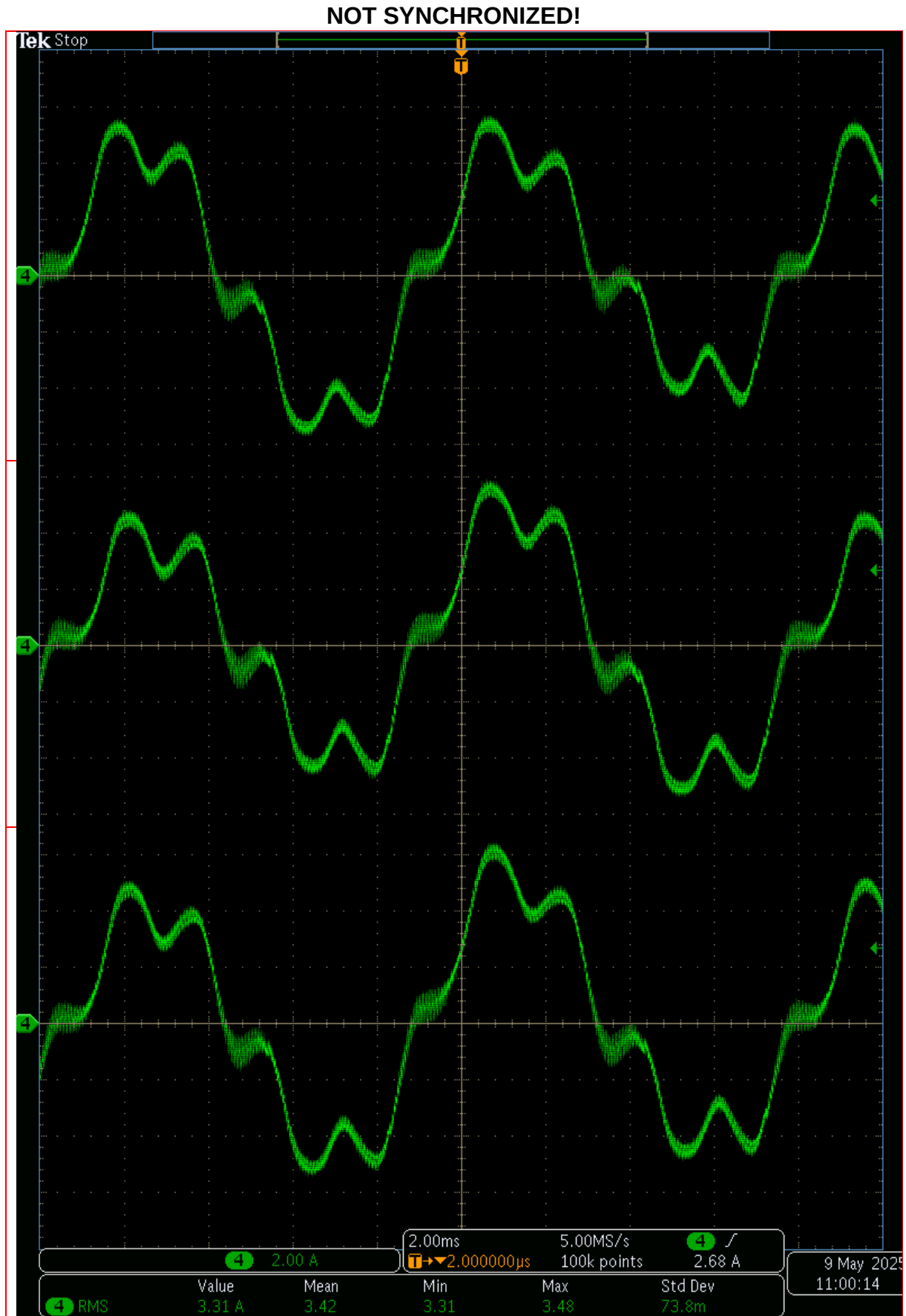


Figure 12. Current waveforms at U (top), V (middle), W (bottom) - 200182100128 (reworked)

6.3. Control

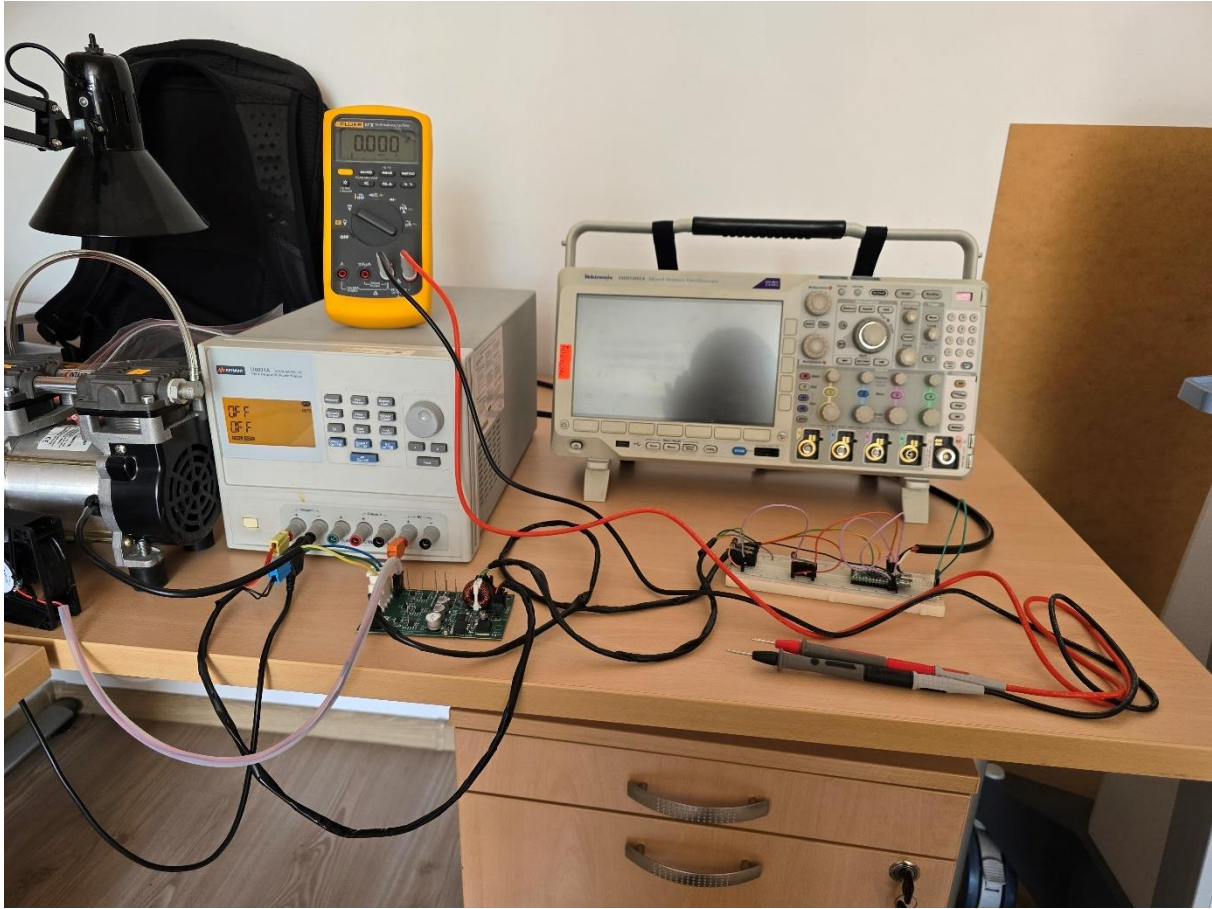


Figure 13. Control test setup

Table 4. Control measurements

Serial number	200182100126	200182100128 (reworked)
Stage 1: Pump/Fan OFF		
$U_{\text{FAULT_RST}}$ [V] (TP13)	0.832	2.558
$U_{\text{ON/OFF}}$ [V] (TP12)	0.000	0.000
U_{FAULT} [V] (TP14)	3.314	3.312
Stage 2: Pump/Fan ON		
$U_{\text{FAULT_RST}}$ [V] (TP13)	0.832	2.558
$U_{\text{ON/OFF}}$ [V] (TP12)	3.314	3.311
U_{FAULT} [V] (TP14)	3.314	3.311

7. Temperature measurements

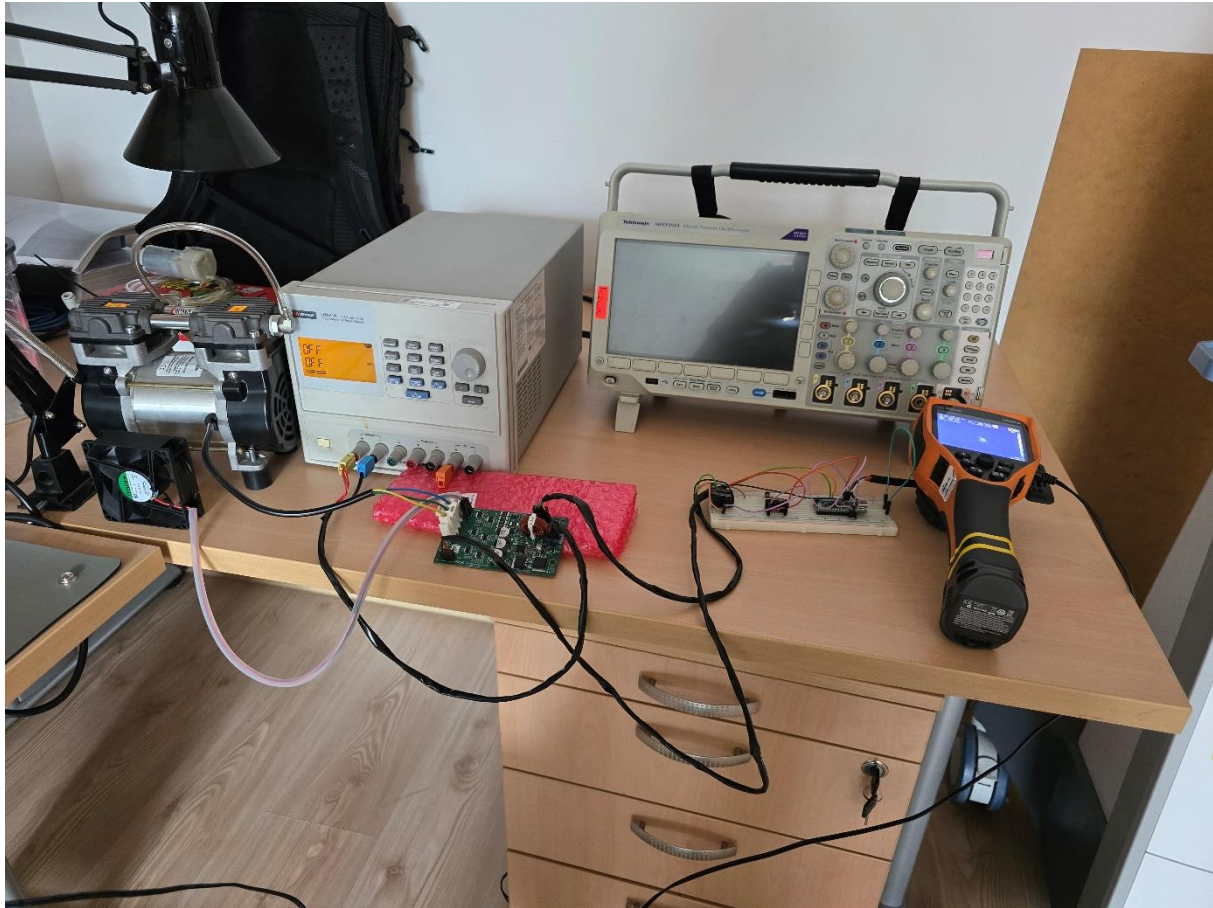


Figure 14. Temperature measurements test setup

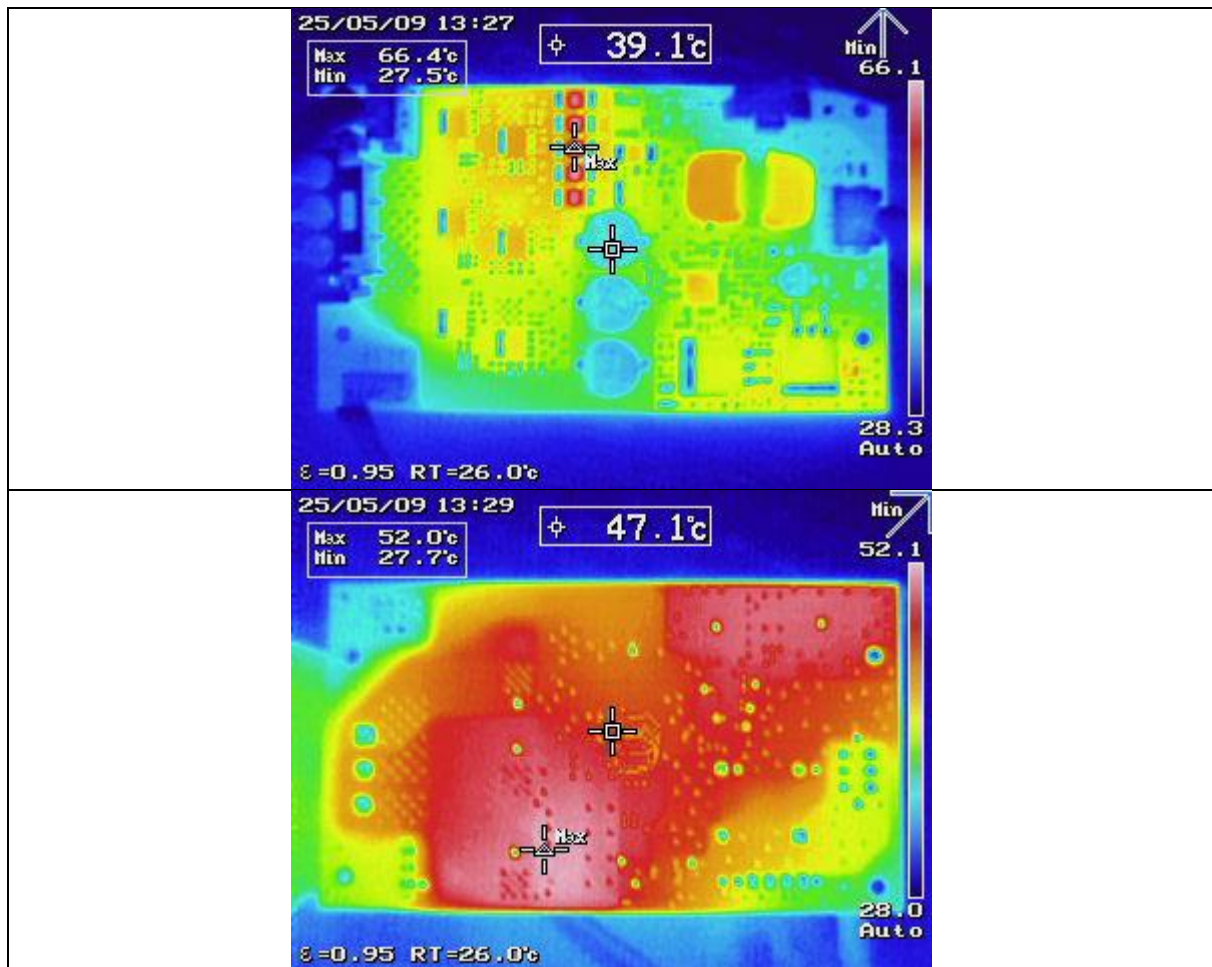


Figure 16. Temperature measurements of 200182100128 (reworked) - top and bottom sides

8. Conclusion

Based on the presented measurements, it can be concluded that certain changes should be made to the PCB for the next revision:

- The inverter's gate resistances should be increased;
- The inverter's snubber resistances should be increased;
- The inverter's snubber capacitances should be decreased;
- The control logic pull-up and pull-down resistances should be decreased;
- The generator connector J5 should be rotated by 180° for easier access.